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(1) Research Program/Project/Study Title: <i>Post-harvest Behaviour of Organically Grown Radish Applied with Different Soil Ameliorants</i>	
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Program or Project has component (*if no proceed to 2*)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☒ Natural/BiologicalSciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☒ CSRDC Crop Science Research Development Center	
(4) Site of implementation			
		Municipality/City:	Mambusao
		Province:	Capiz
		Region:	VI
(5) Classification Research: Development: ☐ Basic ☐ Pilot Testing ☒ Applied ☐ Tech. Promotion/Commercialization		(6) Mode of Implementation ☒ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
(7) Sector/Commodity Vegetable		(8) Discipline/Field of Specialization Agriculture	

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(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.)

The experiment was conducted at the Crop and Soil Science Laboratory of Capiz State University - Burias Campus, Burias, Mambusao, Capiz. It was designed as a single-factor experiment laid out in a Completely Randomized Design (CRD) with three replications. The treatments consisted of different soil ameliorants, namely: T1 - Vermicast, T2 - Vermitea, and T3 - Ordinary Compost. The study recorded the visual quality rating (VQR), shriveling rate, and chemical analysis. The data collected were subjected to the Analysis of Variance (ANOVA) and F-test for Factorial CRD using STAR software.

The result of the study showed the different soil ameliorants significantly influenced the visual quality rating (VQR) of radishes on the twelfth day of storage. There is a gradual increase in shriveling from the eighth day to the twelfth day of storage for all treatments. The pH levels decreased over time regardless of the soil ameliorants applied. The initial and final total soluble solids (TSS) were not affected by the application of the different soil ameliorants.

(9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.)

organic farming, soil ameliorants, vermicompost, compost, radish

(10) Introduction (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews)

Radish (*Raphanus sativus*) is a versatile and nutritious root vegetable widely cultivated across the globe, valued for its crisp texture, peppery flavor, and high nutritional content (Chung et al., 2020). With the increasing demand for organic produce and sustainable farming practices, the cultivation of organically grown radish has gained momentum, driven by the desire to reduce synthetic chemical inputs and promote environmental sustainability (Makoi et al., 2012). Organic radish production aims to enhance soil health, minimize environmental impact, and provide consumers with a healthy and environmentally responsible food choice.

The post-harvest phase is a crucial aspect of radish production, significantly affecting the overall quality, shelf-life, and market value of the crop. Post-harvest losses can occur due to various factors, including mechanical damage, pathogen contamination, and unfavorable storage conditions (Kader, 2002). To mitigate these challenges and improve the post-harvest behavior of organically grown radishes, it is imperative to investigate the potential benefits of different soil ameliorants applied during the cultivation process.

Soil ameliorants, including organic matter and mineral amendments, have been widely recognized for their ability to enhance soil fertility and structure (Zotarelli et al., 2018). The application of soil ameliorants can influence root development, nutrient uptake, and overall crop performance. Previous research has demonstrated the positive effects of soil ameliorants on the yield and quality of various crops, including radishes (Manojlovic et al., 2018). However, there is a dearth of information on how

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different soil ameliorants impact the post-harvest behavior of organically grown radishes.

This study aimed to determine the post-harvest behavior of organically grown sweet corn applied with different soil ameliorants, in terms of visual quality rating, shriveling rate, pH levels, and total soluble solids. By exploring the intricate relationships between soil ameliorants and the post-harvest behavior of radishes, this research contributes to the development of sustainable and environmentally friendly practices for radish production, benefiting both growers and consumers.

Objectives of the Study

In general, the purpose of this study was to determine the post-harvest behaviour of organically grown radishes applied with different soil ameliorants. Specifically, the study aimed to;

1. evaluate the visual quality rating of organically grown radishes after post-harvest storage;
2. determine the shriveling rate of radishes when subjected to different soil ameliorants during post-harvest storage;
3. measure the changes in pH levels of radishes after post-harvest storage when treated with different soil ameliorants;
4. quantify the total soluble solids (TSS) content in organically grown radishes following post-harvest storage with varying soil ameliorants.

(11) Materials and Methods or Methodology (This includes the materials, methods, duration & location of the study. Materials and Methods are headings used in Natural Science researches and methodology is Social Sciences. Presentation of materials may not necessarily be in separate heading but may be incorporated in the specific methods used. Justify paragraphs with main headings in caps.)

The experiment was conducted at the Crop and Soil Science Laboratory of Capiz State University - Burias Campus, Burias, Mambusao, Capiz. It was designed as a single-factor experiment laid out in a Completely Randomized Design (CRD) with three replications. The treatments consisted of different soil ameliorants, namely: T1 - Vermicast, T2 - Vermitea, and T3 - Ordinary Compost. The study recorded the visual quality rating (VQR), shriveling rate, and chemical analysis. The data collected were subjected to the Analysis of Variance (ANOVA) and F-test for Factorial CRD using STAR software. Mean differences among treatments were determined using the Least Significant Difference (LSD).

(12) Results and Discussions (This part contains presentation of results (data) with reference to the table and figure, to be followed by discussion, with support of reviewed literature. Sequence shall be based on the objectives of the study. Tables are formatted without the vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)

Visual Quality Rating

The ANOVA showed that the Visual Quality Rating (VQR) of radish was not significantly affected by the various soil ameliorants, except on the twelfth day of

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storage. There was no significant difference among treatments during storage days 2, 4, 6, 8, and 10. However, on the twelfth day of storage, a significant variation was observed. During this period, Vermicast had the highest visual quality rating (2.55), while radish treated with Vermitea and ordinary compost had the lowest visual quality rating (2). This means that the radish treated with Vermicast was of higher quality than those treated with Vermitea and ordinary compost, which were both interpreted as being close to the limit of edibility.

The visual quality of radishes tends to gradually decline during storage, reflecting the natural aging and degradation processes inherent in harvested radishes (Ramos et al., 2018). These changes can include shifts in color, dehydration, and physiological responses, all of which can negatively impact the visual appeal and marketability of radishes (Borges et al., 2019).

Shriveling Rate

The shriveling rate of radish showed a gradual increase from the eighth day of storage to the twelfth day, regardless of the soil ameliorants applied. This happened regardless of the soil ameliorants that were applied. By the twelfth day, the shriveling rate reached a mark of 4. This means that 50% of the radish area was severely affected by shriveling.

However, the ANOVA test showed that the different soil ameliorants did not have a significant effect on the shriveling rate of radish. This suggests that the application of different soil ameliorants did not cause any change in the shriveling rate of radish.

Shriveling in radishes is primarily the result of moisture loss, which can occur due to respiration, transpiration, and changes in the physiological and biochemical properties of the root tissue. During storage, radishes continue to respire, leading to the release of moisture, which in turn results in the shrinkage of the root tissues. While various factors, such as temperature and humidity, can influence the rate of moisture loss, the application of different treatments, such as the use of soil ameliorants, may also play a role in these changes (Gichimu et al., 2015). In many cases, radishes experience a gradual increase in shriveling as they are subjected to storage conditions (Ahmed et al., 2016). These changes can have implications for the overall appearance and marketability of radishes.

Initial pH and Final pH

The results of the study showed that the initial pH level of radish applied with Vermitea was the highest at 6.35, followed by radish applied with Vermicast at 6.31. The lowest initial pH level was found in radishes applied with ordinary compost at 6.30. Furthermore, the final pH level of radish applied with ordinary compost was the highest at 5.99, while the pH level of radish applied with Vermitea and Vermicast was an average of 5.84. It was found that the pH level of the radish decreased over time from the initial measurement to the final day of storage.

However, the Analysis of Variance revealed that the application of soil

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ameliorants did not significantly affect the initial and final pH levels of radish.

It is well-established that pH fluctuations can naturally occur in harvested produce, with factors such as enzymatic activity and respiration contributing to these changes (Garg and Seth, 2016). In many cases, the pH levels in radishes tend to decrease over time during storage (Mukherjee et al., 2019). This phenomenon can be attributed to the enzymatic and chemical changes occurring within the root tissue.

Initial TSS and Final TSS

The ANOVA shows that the effects of different soil ameliorants on the Total Soluble Solids (TSS) of radish were not significant. This suggests that there were no notable differences observed between the initial and final TSS of radish. The initial TSS mean ranged from 5.27 to 5.67 and the final TSS mean ranged from 5.93 to 6.23.

TSS levels can vary during the post-harvest period due to enzymatic activities, respiration, and other metabolic processes that can impact the overall quality of radishes (Janave et al., 2017). While soil ameliorants are commonly employed to improve soil fertility and structure during cultivation, their influence on the sugar content of radishes may not be substantial. Previous research suggests that in many cases, the application of different soil ameliorants does not significantly affect the initial and final TSS levels of radishes (Vasanthakumari et al., 2015).

(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means, conclusions and recommendations may be split under separate headings if they are long. Justify paragraphs with main headings in caps.)

Conclusion

Based on the results of the study, the following conclusions were drawn;

1. The different soil ameliorants significantly influenced the visual quality rating (VQR) of radishes on the twelfth day of storage.
2. There is a gradual increase in the shriveling of radishes from the eighth day to the twelfth day of storage for all treatments.
3. The pH levels in radishes decreased over time regardless of the soil ameliorants applied.
4. The initial and final total soluble solids (TSS) of radishes were not affected by the application of the different soil ameliorants.

Recommendation

Based on the results of the study, the following conclusions were drawn;

1. Use any of the soil ameliorants in growing radish organically.
2. Conduct studies considering the following:
 - a. storage practices to maintain visual quality over time, such as improved

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- packaging and temperature control.
- b. optimizing storage conditions and handling practices to prevent shriveling, particularly from the eighth to the 12th day of storage.
- c. methods to stabilize pH during the storage of different crops.
- d. develop strategies to maintain the shriveling rate in the storage of different crops.

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indentation with 12-point space in between.)

Ahmed, M., Singh, Z., & Khan, A. S. (2016). Ethylene and 1-MCP regulate low-temperature-induced symptom expression of superficial scald in stored pomegranate fruit. *Postharvest Biology and Technology*, 111, 125-131.

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Appendices

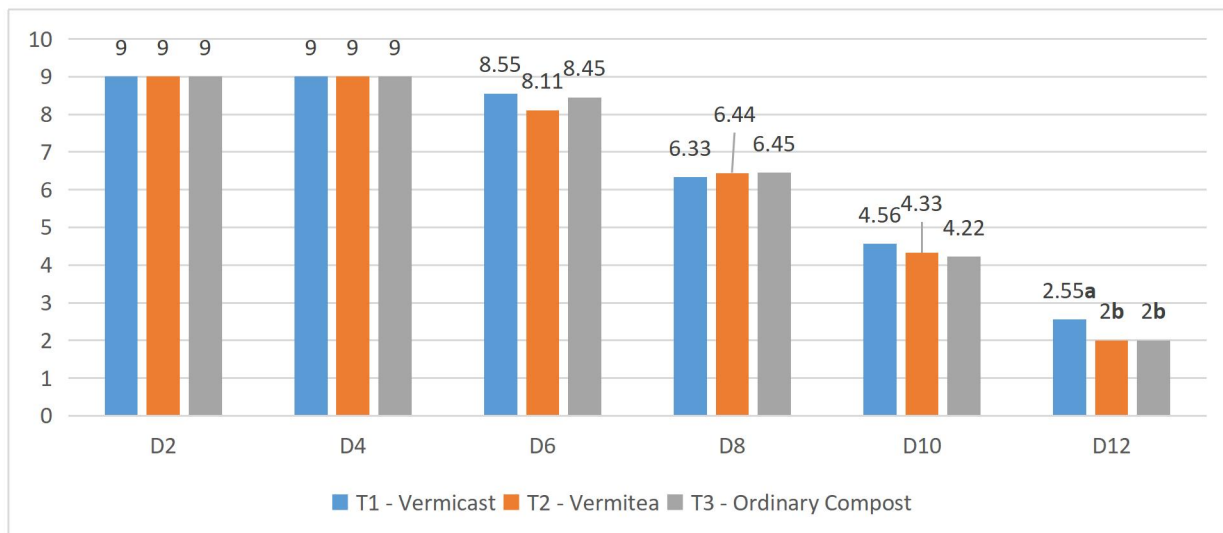


Figure 1. A graphic chart of the Visual Quality Rating (VQR) of radish applied with different soil ameliorants.

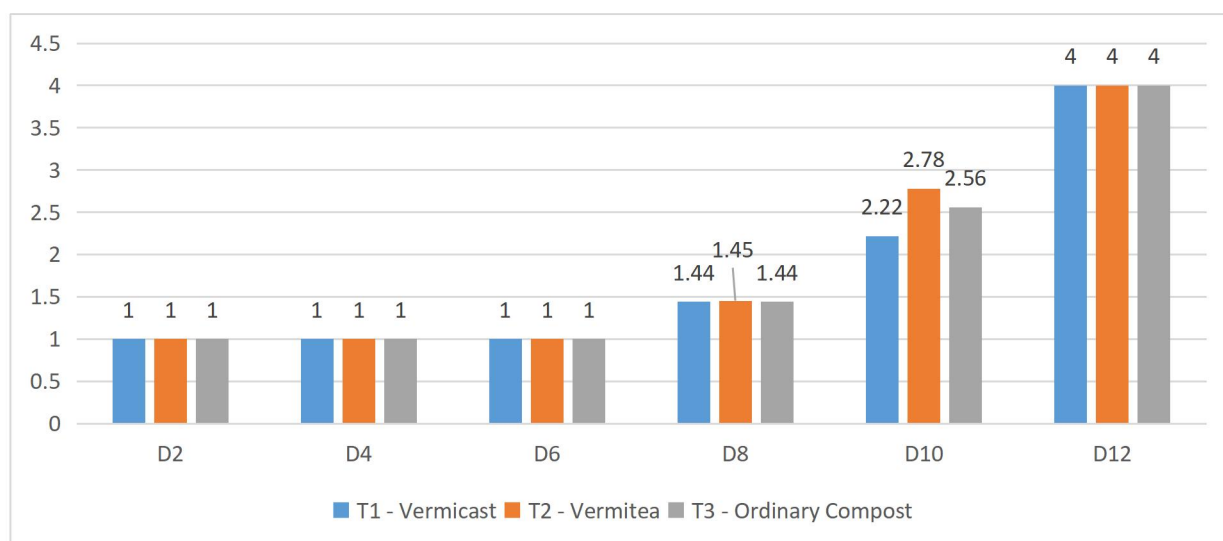


Figure 2. A graphic chart of the Shriveling rate of radish applied with different soil ameliorants.

TREATMENT	Initial pH ^{ns}	Final pH ^{ns}	Initial TSS(Brix) ^{ns}	Final TSS(Brix) ^{ns}
T1 – Vermicast	6.31	5.84	5.67	6.23
T2 – Vermitea	6.35	5.84	5.53	6.13
T3 - Ordinary Compost	6.3	5.99	5.27	5.93

Table 1. Initial pH, Final pH, Initial TSS, and Final TSS of sweet corn applied with different soil ameliorant.